betaSandwich: Internal Tests

Ivan Jacob Agaloos Pesigan

Tests

```
#> test-betaSandwich-beta-sandwich-adf
#> Test passed with 2 successes.
#> Test passed with 2 successes.

#> test-betaSandwich-beta-sandwich-hc

#> Test passed with 2 successes.
#> Test passed with 2 successes.
#> Test passed with 2 successes.
#> Test passed with 2 successes.
#> Test passed with 2 successes.
#> Test passed with 2 successes.
#> Test passed with 2 successes.
#> Test passed with 2 successes.

#> test-betaSandwich-beta-sandwich-methods

#> Call:
#> BetaHC(object = object)
#>
#> Standardized regression slopes with HC3 standard errors:
#>            est     se      t df      p   0.05%   0.5%   2.5%  97.5%  99.5%
#> NARTIC  0.4951 0.0786 6.3025 42 0.0000  0.2172 0.2832 0.3366 0.6537 0.7071
#> PCTGRT  0.3915 0.0818 4.7831 42 0.0000  0.1019 0.1707 0.2263 0.5567 0.6123
#> PCTSUPP 0.2632 0.0855 3.0786 42 0.0037 -0.0393 0.0325 0.0907 0.4358 0.4940
#>         99.95%
#> NARTIC  0.7731
#> PCTGRT  0.6810
#> PCTSUPP 0.5658
#> Call:
#> BetaHC(object = object)
#>
#> Standardized regression slopes with HC3 standard errors:
#>            est     se      t df      p   0.05%   0.5%   2.5%  97.5%  99.5%
#> NARTIC  0.4951 0.0786 6.3025 42 0.0000  0.2172 0.2832 0.3366 0.6537 0.7071
```

```
#> PCTGRT  0.3915 0.0818 4.7831 42 0.0000  0.1019 0.1707 0.2263 0.5567 0.6123
#> PCTSUPP 0.2632 0.0855 3.0786 42 0.0037 -0.0393 0.0325 0.0907 0.4358 0.4940
#>         99.95%
#> NARTIC  0.7731
#> PCTGRT  0.6810
#> PCTSUPP 0.5658
#> Call:
#> BetaN(object = object)
#>
#> Standardized regression slopes with MVN standard errors:
#>            est     se      t df     p   0.05%   0.5%   2.5%  97.5%  99.5%
#> NARTIC  0.4951 0.0759 6.5272 42 0.000  0.2268 0.2905 0.3421 0.6482 0.6998
#> PCTGRT  0.3915 0.0770 5.0824 42 0.000  0.1190 0.1837 0.2360 0.5469 0.5993
#> PCTSUPP 0.2632 0.0747 3.5224 42 0.001 -0.0011 0.0616 0.1124 0.4141 0.4649
#>         99.95%
#> NARTIC  0.7635
#> PCTGRT  0.6640
#> PCTSUPP 0.5276
#> Call:
#> BetaN(object = object)
#>
#> Standardized regression slopes with MVN standard errors:
#>            est     se      t df     p   0.05%   0.5%   2.5%  97.5%  99.5%
#> NARTIC  0.4951 0.0759 6.5272 42 0.000  0.2268 0.2905 0.3421 0.6482 0.6998
#> PCTGRT  0.3915 0.0770 5.0824 42 0.000  0.1190 0.1837 0.2360 0.5469 0.5993
#> PCTSUPP 0.2632 0.0747 3.5224 42 0.001 -0.0011 0.0616 0.1124 0.4141 0.4649
#>         99.95%
#> NARTIC  0.7635
#> PCTGRT  0.6640
#> PCTSUPP 0.5276
#> Call:
#> BetaADF(object = object)
#>
#> Standardized regression slopes with MVN standard errors:
#>            est     se      t df      p   0.05%   0.5%   2.5%  97.5%  99.5%
#> NARTIC  0.4951 0.0674 7.3490 42 0.0000  0.2568 0.3134 0.3592 0.6311 0.6769
#> PCTGRT  0.3915 0.0710 5.5164 42 0.0000  0.1404 0.2000 0.2483 0.5347 0.5830
#> PCTSUPP 0.2632 0.0769 3.4231 42 0.0014 -0.0088 0.0558 0.1081 0.4184 0.4707
#>         99.95%
#> NARTIC  0.7335
#> PCTGRT  0.6426
#> PCTSUPP 0.5353
#> Call:
#> BetaADF(object = object)
#>
#> Standardized regression slopes with MVN standard errors:
```

```
#>            est     se      t df      p   0.05%   0.5%   2.5%  97.5%  99.5%
#> NARTIC  0.4951 0.0674 7.3490 42 0.0000  0.2568 0.3134 0.3592 0.6311 0.6769
#> PCTGRT  0.3915 0.0710 5.5164 42 0.0000  0.1404 0.2000 0.2483 0.5347 0.5830
#> PCTSUPP 0.2632 0.0769 3.4231 42 0.0014 -0.0088 0.0558 0.1081 0.4184 0.4707
#>         99.95%
#> NARTIC  0.7335
#> PCTGRT  0.6426
#> PCTSUPP 0.5353
#> Call:
#> BetaHC(object = object)
#>
#> Standardized regression slopes with HC3 standard errors:
#>           est     se       t df p  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC 0.7622 0.0645 11.8222 44 0 0.5349 0.5886 0.6322 0.8921 0.9357 0.9895
#> Call:
#> BetaHC(object = object)
#>
#> Standardized regression slopes with HC3 standard errors:
#>           est     se       t df p  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC 0.7622 0.0645 11.8222 44 0 0.5349 0.5886 0.6322 0.8921 0.9357 0.9895
#> Call:
#> BetaN(object = object)
#>
#> Standardized regression slopes with MVN standard errors:
#>           est     se       t df p  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC 0.7622 0.0618 12.3341 44 0 0.5443 0.5958 0.6376 0.8867 0.9285   0.98
#> Call:
#> BetaN(object = object)
#>
#> Standardized regression slopes with MVN standard errors:
#>           est     se       t df p  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC 0.7622 0.0618 12.3341 44 0 0.5443 0.5958 0.6376 0.8867 0.9285   0.98
#> Call:
#> BetaADF(object = object)
#>
#> Standardized regression slopes with MVN standard errors:
#>           est     se      t df p  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC 0.7622 0.0604 12.625 44 0 0.5493 0.5996 0.6405 0.8838 0.9247  0.975
#> Call:
#> BetaADF(object = object)
#>
#> Standardized regression slopes with MVN standard errors:
#>           est     se      t df p  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC 0.7622 0.0604 12.625 44 0 0.5493 0.5996 0.6405 0.8838 0.9247  0.975
#> Test passed with 1 success.
```

```
#> test-betaSandwich-beta-sandwich-mvn

#> Test passed with 2 successes.
#> Test passed with 2 successes.

#> test-betaSandwich-beta-sandwich-std

#> Test passed with 13 successes.

#> test-betaSandwich-beta-star-select

#> Test passed with 8 successes.

#> test-betaSandwich-delta-r-sq-sandwich

#> Test passed with 3 successes.
#> Test passed with 3 successes.
#> Test passed with 3 successes.

#> test-betaSandwich-diff-beta-sandwich

#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.

#> test-betaSandwich-es-sandwich-methods

#> Test passed with 811 successes.

#> test-betaSandwich-f-sq-sandwich

#> Test passed with 3 successes.
#> Test passed with 3 successes.
#> Test passed with 3 successes.

#> test-betaSandwich-jac-beta-star-sel-wrt-theta-star
```

```
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.

#> test-betaSandwich-jac-delta-r-sq-wrt-theta-star

#> Test passed with 2 successes.
#> Test passed with 2 successes.
#> Test passed with 2 successes.

#> test-betaSandwich-jac-f-sq-wrt-theta-star

#> Test passed with 2 successes.
#> Test passed with 2 successes.
#> Test passed with 2 successes.

#> test-betaSandwich-jac-p-cor-sq-wrt-theta-star

#> Test passed with 2 successes.
#> Test passed with 2 successes.
#> Test passed with 2 successes.

#> test-betaSandwich-jac-r-sq-wrt-theta-star

#> Test passed with 2 successes.
#> Test passed with 2 successes.
#> Test passed with 2 successes.

#> test-betaSandwich-jac-s-cor-wrt-theta-star

#> Test passed with 2 successes.
#> Test passed with 2 successes.
#> Test passed with 2 successes.

#> test-betaSandwich-jac-vech-sigma-wrt-theta-star

#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.

#> test-betaSandwich-p-cor-sandwich

#> Test passed with 3 successes.
#> Test passed with 3 successes.
#> Test passed with 3 successes.

#> test-betaSandwich-r-sq-sandwich-mc
```

```
#> Test passed with 3 successes.
#> Test passed with 3 successes.
#> Test passed with 3 successes.

#> test-betaSandwich-r-sq-sandwich-se

#> Test passed with 3 successes.

#> test-betaSandwich-r-sq-sandwich

#> Test passed with 2 successes.
#> Test passed with 2 successes.
#> Test passed with 2 successes.
#> Test passed with 2 successes.
#> Test passed with 2 successes.
#> Test passed with 2 successes.

#> test-betaSandwich-s-cor-sandwich

#> Test passed with 3 successes.
#> Test passed with 3 successes.
#> Test passed with 3 successes.

#> test-zzz-coverage

#>               beta1    beta2    beta3 sigmasq   sigmax1x1  sigmax2x1  sigmax3x1
#> sigmaysq   909.1981 257.2976 276.0367       1 0.007091036 0.03637752 0.01896371
#> sigmayx1  3507.1691 471.2058 510.5430       0 0.084208291 0.21599726 0.11260003
#> sigmayx2   471.2058 333.2295 150.9121       0 0.000000000 0.08420829 0.00000000
#> sigmayx3   510.5430 150.9121 554.4386       0 0.000000000 0.00000000 0.08420829
#> sigmax1x1    0.0000   0.0000   0.0000       0 1.000000000 0.00000000 0.00000000
#> sigmax2x1    0.0000   0.0000   0.0000       0 0.000000000 1.00000000 0.00000000
#> sigmax3x1    0.0000   0.0000   0.0000       0 0.000000000 0.00000000 1.00000000
#> sigmax2x2    0.0000   0.0000   0.0000       0 0.000000000 0.00000000 0.00000000
#> sigmax3x2    0.0000   0.0000   0.0000       0 0.000000000 0.00000000 0.00000000
#> sigmax3x3    0.0000   0.0000   0.0000       0 0.000000000 0.00000000 0.00000000
#>            sigmax2x2 sigmax3x2  sigmax3x3
#> sigmaysq  0.04665482 0.0486426 0.01267877
#> sigmayx1  0.00000000 0.0000000 0.00000000
#> sigmayx2  0.21599726 0.1126000 0.00000000
#> sigmayx3  0.00000000 0.2159973 0.11260003
#> sigmax1x1 0.00000000 0.0000000 0.00000000
#> sigmax2x1 0.00000000 0.0000000 0.00000000
#> sigmax3x1 0.00000000 0.0000000 0.00000000
#> sigmax2x2 1.00000000 0.0000000 0.00000000
#> sigmax3x2 0.00000000 1.0000000 0.00000000
#> sigmax3x3 0.00000000 0.0000000 1.00000000
#>               beta1    beta2    beta3 sigmasq
#> sigmaysq   909.1981 257.2976 276.0367       1
```

```
#> sigmayx1  3507.1691 471.2058 510.5430       0
#> sigmayx2   471.2058 333.2295 150.9121       0
#> sigmayx3   510.5430 150.9121 554.4386       0
#> sigmax1x1    0.0000   0.0000   0.0000       0
#> sigmax2x1    0.0000   0.0000   0.0000       0
#> sigmax3x1    0.0000   0.0000   0.0000       0
#> sigmax2x2    0.0000   0.0000   0.0000       0
#> sigmax3x2    0.0000   0.0000   0.0000       0
#> sigmax3x3    0.0000   0.0000   0.0000       0
#>               beta1    beta2    beta3       rsq   sigmax1x1  sigmax2x1
#> sigmaysq  1130.1037 319.8125 343.1047 -126.0843 0.008813928 0.04521608
#> sigmayx1  3507.1691 471.2058 510.5430    0.0000 0.084208291 0.21599726
#> sigmayx2   471.2058 333.2295 150.9121    0.0000 0.000000000 0.08420829
#> sigmayx3   510.5430 150.9121 554.4386    0.0000 0.000000000 0.00000000
#> sigmax1x1    0.0000   0.0000   0.0000    0.0000 1.000000000 0.00000000
#> sigmax2x1    0.0000   0.0000   0.0000    0.0000 0.000000000 1.00000000
#> sigmax3x1    0.0000   0.0000   0.0000    0.0000 0.000000000 0.00000000
#> sigmax2x2    0.0000   0.0000   0.0000    0.0000 0.000000000 0.00000000
#> sigmax3x2    0.0000   0.0000   0.0000    0.0000 0.000000000 0.00000000
#> sigmax3x3    0.0000   0.0000   0.0000    0.0000 0.000000000 0.00000000
#>            sigmax3x1  sigmax2x2  sigmax3x2 sigmax3x3
#> sigmaysq  0.02357128 0.05799042 0.06046117 0.0157593
#> sigmayx1  0.11260003 0.00000000 0.00000000 0.0000000
#> sigmayx2  0.00000000 0.21599726 0.11260003 0.0000000
#> sigmayx3  0.08420829 0.00000000 0.21599726 0.1126000
#> sigmax1x1 0.00000000 0.00000000 0.00000000 0.0000000
#> sigmax2x1 0.00000000 0.00000000 0.00000000 0.0000000
#> sigmax3x1 1.00000000 0.00000000 0.00000000 0.0000000
#> sigmax2x2 0.00000000 1.00000000 0.00000000 0.0000000
#> sigmax3x2 0.00000000 0.00000000 1.00000000 0.0000000
#> sigmax3x3 0.00000000 0.00000000 0.00000000 1.0000000
#>               beta1    beta2    beta3       rsq
#> sigmaysq  1130.1037 319.8125 343.1047 -126.0843
#> sigmayx1  3507.1691 471.2058 510.5430    0.0000
#> sigmayx2   471.2058 333.2295 150.9121    0.0000
#> sigmayx3   510.5430 150.9121 554.4386    0.0000
#> sigmax1x1    0.0000   0.0000   0.0000    0.0000
#> sigmax2x1    0.0000   0.0000   0.0000    0.0000
#> sigmax3x1    0.0000   0.0000   0.0000    0.0000
#> sigmax2x2    0.0000   0.0000   0.0000    0.0000
#> sigmax3x2    0.0000   0.0000   0.0000    0.0000
#> sigmax3x3    0.0000   0.0000   0.0000    0.0000
#> Test passed with 1 success.
#> [[1]]
#> [[1]][[1]]
#> [[1]][[1]]$value
```

```
#> [[1]][[1]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[1]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[2]]
#> [[1]][[2]]$value
#> [[1]][[2]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[2]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[3]]
#> [[1]][[3]]$value
#> [[1]][[3]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[3]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[4]]
#> [[1]][[4]]$value
#> [[1]][[4]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[4]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[5]]
#> [[1]][[5]]$value
#> [[1]][[5]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[5]]$visible
```

```
#> [1] TRUE
#>
#>
#> [[1]][[6]]
#> [[1]][[6]]$value
#> [[1]][[6]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[6]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[7]]
#> [[1]][[7]]$value
#> [[1]][[7]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[7]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[8]]
#> [[1]][[8]]$value
#> [[1]][[8]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[8]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[9]]
#> [[1]][[9]]$value
#> [[1]][[9]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[9]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[10]]
#> [[1]][[10]]$value
```

```
#> [[1]][[10]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[10]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[11]]
#> [[1]][[11]]$value
#> [[1]][[11]]$value[[1]]
#> NULL
#>
#>
#> [[1]][[11]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[12]]
#> [[1]][[12]]$value
#> [[1]][[12]]$value[[1]]
#> NULL
#>
#>
#> [[1]][[12]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[13]]
#> [[1]][[13]]$value
#> [[1]][[13]]$value[[1]]
#> NULL
#>
#>
#> [[1]][[13]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[14]]
#> [[1]][[14]]$value
#> [[1]][[14]]$value[[1]]
#> NULL
#>
#>
#> [[1]][[14]]$visible
```

```
#> [1] TRUE
#>
#>
#> [[1]][[15]]
#> [[1]][[15]]$value
#> [[1]][[15]]$value[[1]]
#> NULL
#>
#>
#> [[1]][[15]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[16]]
#> [[1]][[16]]$value
#> [[1]][[16]]$value[[1]]
#> NULL
#>
#>
#> [[1]][[16]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[17]]
#> [[1]][[17]]$value
#> [[1]][[17]]$value[[1]]
#> NULL
#>
#>
#> [[1]][[17]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[18]]
#> [[1]][[18]]$value
#> [[1]][[18]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[18]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[19]]
#> [[1]][[19]]$value
```

```
#> [[1]][[19]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[19]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[20]]
#> [[1]][[20]]$value
#> [[1]][[20]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[20]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[21]]
#> [[1]][[21]]$value
#> [[1]][[21]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[21]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[22]]
#> [[1]][[22]]$value
#> [[1]][[22]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[22]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[23]]
#> [[1]][[23]]$value
#> [[1]][[23]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[23]]$visible
```

```
#> [1] TRUE
```

Environment

```
ls()

#> [1] "nas1982" "root"
```

Class

```
#> [[1]]
#> [1] "data.frame"
#>
#> [[2]]
#> [1] "root_criterion"
```

References

Pesigan, I. J. A., Sun, R. W., & Cheung, S. F. (2023). betaDelta and betaSandwich: Confidence intervals for standardized regression coefficients in R. *Multivariate Behavioral Research*, *58*(6), 1183–1186. https://doi.org/10.1080/00273171.2023.2201277

R Core Team. (2026). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. Vienna, Austria. https://www.R-project.org/